

NVIDIA FY2025 Q4 earnings call transcript

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Nvidia (NASDAQ: NVDA)

Q4 2025 Earnings Call

Feb 26, 2025 , 5:00 p.m. ET

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Prepared Remarks:

Operator

Good afternoon. My name is Krista, and I will be your conference operator today. At this time, I would like to welcome everyone to NVIDIA's fourth-quarter earnings call. All lines have been placed on mute to prevent any background noise.

After the speakers' remarks, there will be a question-and-answer session. [Operator instructions] Thank you. Stewart Stecker, you may begin your conference.

Stewart Stecker -- Senior Director, Investor Relations

Thank you. Good afternoon, everyone, and welcome to NVIDIA's conference call for the fourth quarter of fiscal 2025. With me today from NVIDIA are Jensen Huang, president and chief executive officer; and Colette Kress, executive vice president and chief financial officer. I'd like to remind you that our call is being webcast live on NVIDIA's Investor Relations website.

The webcast will be available for replay until the conference call to discuss our financial results for the first quarter of fiscal 2026. The content of today's call is NVIDIA's property. It can't be reproduced or transcribed without prior written consent. During this call, we may make forward-looking statements based on current expectations.

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These are subject to a number of significant risks and uncertainties, and our actual results may differ materially. For a discussion of factors that could affect our future financial results and business, please refer to the disclosure in today's earnings release, our most recent forms 10-K and 10-Q, and the reports that we may file on Form 8-K with the Securities and Exchange Commission. All our statements are made as of today, February 26, 2025, based on information currently available to us. Except as required by law, we assume no obligation to update any such statements.

During this call, we will discuss non-GAAP financial measures. Confine a reconciliation of these non-GAAP financial measures to GAAP financial measures in our CFO commentary, which is posted on our website. With that, let me turn the call over to Colette.

Colette M. Kress -- Chief Financial Officer, Executive Vice President

Thanks, Stewart. Q4 was another record quarter. Revenue of \$39.3 billion was up 12% sequentially and up 78% year on year and above our outlook of \$37.5 billion. For fiscal 2025 revenue was \$130.5 billion, up 114% from the prior year.

Let's start with Data Center. Data center revenue for fiscal 2025 was \$115.2 billion, more than doubling from the prior year. In the fourth quarter, Data Center revenue of \$35.6 billion was a record, up 16% sequentially and 93% year on year as the Blackwell ramp commenced and Hopper 200 continued sequential growth. In Q4, Blackwell sales exceeded our expectations.

We delivered \$11 billion of Blackwell revenue to meet strong demand. This is the fastest product ramp in our company's history, unprecedented in its speed and scale. Blackwell production is in full-year across multiple configurations, and we are increasing supply quickly expanding customer adoption. Our Q4 Data Center compute revenue jumped 18% sequentially and over 2x year on year.

Customers are racing to scale infrastructure to train the next generation of cutting-edge models and unlock the next level of AI capabilities. With Blackwell, it will be common for these clusters to start with 100,000 GPUs or more. Shipments have already started for multiple infrastructures of this size. Post-training and model customization are fueling demand for NVIDIA infrastructure and software as developers and enterprises leverage techniques such as fine-tuning reinforcement learning and distillation to tailor models for domain-specific use cases.

Hugging Face alone hosts over 90,000 derivatives freighted from the Llama foundation model. The scale of post-training and model customization is massive and can collectively demand orders of magnitude, more compute than pretraining. Our inference demand is accelerating, driven by test time scaling and new reasoning models like OpenAI's o3, DeepSeek-R1, and Grok 3. Long-thinking reasoning AI can require 100x more compute per task compared to one-shot inferences.

Blackwell was architected for reasoning AI inference. Blackwell supercharges reasoning AI models with up to 25x higher token throughput and 20x lower cost versus Hopper 100. Its revolutionary transformer engine is built for LLM and mixture of experts inference. And its NVLink Domain delivers 14x the throughput of PCIe Gen 5, ensuring the response time, throughput, and cost efficiency needed to tackle the growing complexity of infants of scale.

Companies across industries are tapping into NVIDIA's whole STAG inference platform to boost performance and slash costs. Now, tripled inference throughput and cut costs by 66%, using NVIDIA TensorRT for its screenshot feature. Perplexity sees 435 million monthly queries and reduced its inference costs 3x with NVIDIA Triton Inference Server and TensorRT-LLM. Microsoft

Bing achieved a 5x speed up at major TCO savings for visual search across billions of images with NVIDIA, TensorRT, and acceleration libraries.

Blackwell has great demand for inference. Many of the early GB200 deployments are earmarked for inference, a first for a new architecture. Blackwell addresses the entire AI market from pretraining, post-training to inference across cloud, to on-premise, to enterprise. CUDA's programmable architecture accelerates every AI model and over 4,400 applications, ensuring large infrastructure investments against obsolescence in rapidly evolving markets.

Our performance and pace of innovation is unmatched. We're driven to a 200x reduction in inference costs in just the last two years. We delivered the lowest TCO and the highest ROI. And full stack optimizations for NVIDIA and our large ecosystem, including 5.9 million developers continuously improve our customers' economics.

In Q4, large CSPs represented about half of our data center revenue. and these sales increased nearly 2x year on year. Large CSPs were some of the first to stand up Blackwell with Azure, GCP, AWS, and OCI bringing GB200 systems to cloud regions around the world to meet certain surging customer demand for AI. Regional cloud hosting NVIDIA GPUs increased as a percentage of data center revenue, reflecting continued AI factory build-outs globally and rapidly rising demand for AI reasoning models and agents.

We've launched a 100,000 GB200 cluster-based incidents with NVLink Switch and Quantum 2 InfiniBand. Consumer Internet revenue grew 3x year on year, driven by an expanding set of generative AI and deep learning use cases. These include recommender systems, vision, language understanding, synthetic data generation search, and agentic AI. For example, xAI is adopting the GB200 to train and inference its next generation of Grok AI models.

Meta's cutting-edge Andromeda advertising engine runs on NVIDIA's Grace Hopper Superchip serving vast quantities of ads across Instagram, Facebook applications. Andromeda harnesses Grace Hopper's fast interconnect and large memory to boost inference throughput by 3x, enhance ad personalization, and deliver meaningful jumps in monetization and ROI. Enterprise revenue increased nearly 2x year on accelerating demand for model fine-tuning, RAG, and agentic AI workflows, and GPU-accelerated data processing. We introduced NVIDIA Llama Nemotron model family NIMs to help developers create and deploy AI agents across a range of applications including customer support, fraud detection, and product supply chain and inventory management.

Leading AI agent platform providers, including SAP and ServiceNow, are among the first to use new models. Health care leaders IQVIA, Illumina, Mayo Clinic, and Arc Institute are using NVIDIA AI to speed drug discovery enhanced genomic research and Pioneer advanced healthcare services with generative and agentic AI. As AI expands beyond the digital world, NVIDIA infrastructure and software platforms are increasingly being adopted to power robotics and physical AI development. One of the early and largest robotics applications and autonomous vehicles where virtually, every AV company is developing on NVIDIA in the data center, the car, or both.

NVIDIA's automotive vertical revenue is expected to grow to approximately \$5 billion this fiscal year. At CES, Hyundai Motor Group announced it is adopting NVIDIA technologies to accelerate AV and robotics development and smart factory initiatives. Vision transformers, self-supervised learning, multimodal sensor fusion, and high-fidelity simulation are driving breakthroughs in AV development and will require 10x more compute. At CES, we announced the NVIDIA COSMO World Foundation model platform.

Just as language, foundation models have revolutionized Language AI, Cosmos is a physical AI to revolutionize robotics. The robotics and automotive companies, including ridesharing giant Uber, are among the first to adopt the platform. From a geographic perspective, sequential growth in our Data Center revenue was strongest in the U.S., driven by the initial ramp up Blackwell. Countries across the globe are building their AI ecosystem as demand for compute

infrastructure is surging.

France's EUR 100 billion AI investment and the EU's EUR 200 billion invest AI initiatives offer a glimpse into the build-out to set redefined global AI infrastructure in the coming years. Now, as a percentage of total Data Center revenue, data center sales in China remained well below levels seen on the onset of export controls. Absent any change in regulations, we believe that China shipments will remain roughly at the current percentage. The market in China for data center solutions remains very competitive.

We will continue to comply with export controls while serving our customers. Networking revenue declined 3% sequentially. Our networking attached to GPU compute systems is robust at over 75%. We are transitioning from small NVLink 8 with InfiniBand, to large NVLink 72 with Spectrum-X.

Spectrum-X and NVLink Switch revenue increased and represents a major new growth vector. We expect networking to return to growth in Q1. AI requires a new class of networking. NVIDIA offers NVLink Switch systems for scale-up compute.

For scale-out, we offer quantum incentive for HPC supercomputers and Spectrum X for Ethernet environments. Spectrum-X enhances the Ethernet for AI computing and has been a huge success. Microsoft Azure, OCI, CoreWeave, and others are building large AI factories with Spectrum-X. The first Stargate data centers will use Spectrum-X.

Yesterday, Cisco announced integrating Spectrum-X into their networking portfolio to help enterprises build AI infrastructure. With its large enterprise footprint and global reach, Cisco will bring NVIDIA Ethernet to every industry. Now, moving to gaming and ARPCs. Gaming revenue of \$2.5 billion decreased 22% sequentially and 11% year on year.

Full-year revenue of \$11.4 billion increased 9% year on year, and demand remains strong throughout the holiday. However, Q4 shipments were impacted by supply constraints. We expect strong sequential growth in Q1 as supply increases. The new GeForce RTX 50 Series desktop and laptop GPUs are here.

Build for gamers, creators, and developers they fuse AI and graphics redefining visual computing, powered by the Blackwell architecture, fifth-generation Tensor cores, and fourth-generation RT cores and featuring UQ's400AI tops. These GPUs deliver a 2x performance leap and new AI-driven rendering including neuro shaders, digital human technologies, geometry, and lighting. The new DLSS 4 boost frame rates up to 8x with AI-driven frame generation, turning one rendered frame into three. It also features the industry's first real-time application of transformer models packing 2x more parameters and 4x to compute for unprecedented visual fidelity.

We also announced a wave of GeForce Blackwell laptop GPUs with new NVIDIA Max-Q technology that extends battery life by up to an incredible 40%. These laptops will be available starting in March from the world's top manufacturers. Moving to our professional visualization business. Revenue of \$511 million was up 5% sequentially and 10% year on year.

Full-year revenue of \$1.9 billion increased 21% year on year. Key industry verticals driving demand include automotive and healthcare. NVIDIA Technologies and generative AI are reshaping design, engineering, and simulation workloads. Increasingly, these technologies are being leveraged in leading software platforms from ANSYS, Cadence, and Siemens fueling demand for NVIDIA RTX workstations.

Now, moving to Automotive. Revenue was a record \$570 million, up 27% sequentially and up 103% year on year. Full-year revenue of \$1.7 billion increased 5% year on year. Strong growth was driven by the continued ramp in autonomous vehicles, including cars and robotaxis.

At CES, we announced Toyota, the world's largest automaker will build its next-generation

vehicles on NVIDIA Orin running the safety-certified NVIDIA DriveOS. We announced Aurora and Continental will deploy driverless trucks at scale powered by NVIDIA Drive Thor. Finally, our end-to-end autonomous vehicle platform NVIDIA Drive Hyperion has passed industry safety assessments like TUV SUD and TUV Rheinland, two of the industry's foremost authorities for automotive-grade safety and cybersecurity. NVIDIA is the first AV platform that received a comprehensive set of third-party assessments.

OK. Moving to the rest of the P&L. GAAP gross margin was 73% and non-GAAP gross margin was 73.5% and down sequentially as expected with our first deliveries of the Blackwell architecture. As discussed last quarter, Blackwell is a customizable AI infrastructure with several different types of NVIDIA build chips multiple networking options, and for air and liquid-cooled data center.

We exceeded our expectations in Q4 in ramping Blackwell, increasing system availability, providing several configurations to our customers. As Blackwell ramps, we expect gross margins to be in the low 70s. Initially, we are focused on expediting the manufacturing of Blackwell systems to meet strong customer demand as they race to build out Blackwell infrastructure. When fully ramped, we have many opportunities to improve the cost, and gross margin will improve and return to the mid-70s, late this fiscal year.

Sequentially, GAAP operating expenses were up 9% and non-GAAP operating expenses were 11%, reflecting higher engineering development costs and higher compute and infrastructure costs for new product introductions. In Q4, we returned \$8.1 billion to shareholders in the form of share repurchases and cash dividends. Let me turn to the outlook in the first quarter. Total revenue is expected to be \$43 billion, plus or minus 2%.

Continuing with its strong demand, we expect a significant ramp of Blackwell in Q1. We expect sequential growth in both Data Center and Gaming. Within Data Center, we expect sequential growth from both compute and networking. GAAP and non-GAAP gross margins are expected to be 70.6% and 71%, respectively, plus or minus 50 basis points.

GAAP and non-GAAP operating expenses are expected to be approximately \$5.2 billion and \$3.6 billion, respectively. We expect full-year fiscal year '26 operating expenses to grow to be in the mid-30s. GAAP and non-GAAP other incoming expenses are expected to be an income of approximately \$400 million. excluding gains and losses from nonmarketable and publicly held equity securities.

GAAP and non-GAAP tax rates are expected to be 17%, plus or minus 1%, excluding any discrete items. Further financial details are included in the CFO commentary and other information available on our IR website, including a new financial information AI agent. In closing, let me highlight upcoming events for the financial community. We will be at the TD Cowen Health Care Conference in Boston on March 3 and at the Morgan Stanley Technology, Media, and Telecom Conference in San Francisco on March 5.

Please join us for our Annual GTC conference starting Monday, March 17 in San Jose, California. Jensen will deliver a news-packed keynote on March 18, and we will host a Q&A session for our financial analysts for the next day, March 19. We look forward to seeing you at these events. Our earnings call to discuss the results for our first quarter of fiscal 2026 is scheduled for May 28, 2025.

We are going to open up the call, operator, to questions. If you could start that, that would be great.

Questions & Answers:

Operator

Thank you. [Operator instructions] And your first question comes from C.J. Muse with Cantor

Fitzgerald. Please go ahead.

C.J. Muse -- Analyst

Yeah. Good afternoon. Thank you for taking the question. I guess for me, Jensen, as test-time compute and reinforcement learning shows such promise, we're clearly seeing an increasing blurring of the lines between training and inference.

What does this mean for the potential future of potentially inference dedicated clusters? And how do you think about the overall impact to NVIDIA and your customers? Thank you.

Jensen Huang -- President and Chief Executive Officer

Yeah, I appreciate that, C.J. There are now multiple scaling laws. There's the pre-training scaling law, and that's going to continue to scale because we have multimodality, we have data that came from reasoning that are now used to do pretraining. And then the second is post-training skilling, using reinforcement learning human feedback, reinforcement learning AI feedback, reinforcement learning, verifiable rewards.

The amount of computation you use for post-training is actually higher than pretraining. And it's kind of sensible in the sense that you could, while you're using reinforcement learning, generate an enormous amount of synthetic data or synthetically generated tokens. AI models are basically generating tokens to train AI models. And that's post-trade.

And the third part, this is the part that you mentioned is test-time compute or reasoning, long thinking, inference scaling. They're all basically the same ideas. And there, you have a chain of thought, you have search. The amount of tokens generated the amount of inference compute needed is already 100x more than the one-shot examples and the one-shot capabilities of large language models in the beginning.

And that's just the beginning. This is just the beginning. The idea that the next generation could have thousands of times and even, hopefully, extremely thoughtful and simulation-based and search-based models that could be hundreds of thousands, millions of times more compute than today is in our future. And so, the question is, how do you design such an architecture? Some of it -- some of the models are auto-regressive.

Some of the models are diffusion-based. Some of it -- some of the times you want your data center to have disaggregated inference. Sometimes, it is compacted. And so, it's hard to figure out what is the best configuration of a data center, which is the reason why NVIDIA's architecture is so popular.

We run every model. We are great at training. The vast majority of our compute today is actually inference and Blackwell takes all of that to a new level. We designed Blackwell with the idea of reasoning models in mind.

And when you look at training, it's many times more performing. But what's really amazing is for long thinking test time scaling, reasoning AI models were tens of times faster, 25x higher throughput. And so, Blackwell is going to be incredible across the board. And when you have a data center that allows you to configure and use your data center based on are you doing more pretraining now, post-training now, or scaling out your inference, our architecture is fungible and easy to use in all of those different ways.

And so, we're seeing, in fact, much, much more concentration of a unified architecture than ever before.

Operator

Your next question comes from the line of Joe Moore with JPMorgan. Please go ahead.

Joe Moore -- JPMorgan Chase and Company -- Analyst

Morgan Stanley, actually. Thank you. I wonder if you could talk about GB200 at CES, you sort of talked about the complexity of the rack-level systems and the challenges you have. And then as you said in the prepared remarks, we've seen a lot of general availability.

Where are you in terms of that ramp? Are there still bottlenecks to consider at a systems level above and beyond the chip level? And just have you maintained your enthusiasm for the NVL72 platforms?

Jensen Huang -- President and Chief Executive Officer

Well, I'm more enthusiastic today than I was at CES. And the reason for that is because we've shipped a lot more since CES. We have some 350 plants manufacturing the 1.5 million components that go into each one of the Blackwell racks, Grace Blackwell racks. Yes, it's extremely complicated.

And we successfully and incredibly ramped up Grace Blackwell, delivering some \$11 billion of revenues last quarter. We're going to have to continue to scale as demand is quite high, and customers are anxious and impatient to get their Blackwell systems. You've probably seen on the web, a fair number of celebrations about Grace Blackwell systems coming online and we have them, of course. We have a fairly large installation of Grace Blackwell for our own engineering and our own design teams and software teams.

CoreWeave has now been quite public about the successful bring-up of theirs. Microsoft has, of course, OpenAI has, and you're starting to see many come online. And so, I think the answer to your question is nothing is easy about what we're doing, but we're doing great, and all of our partners are doing great.

Operator

Your next question comes from the line of Vivek Arya with Bank of America Securities. Please go ahead.

Vivek Arya -- Analyst

Thank you for taking my question. Colette if you wouldn't mind confirming if Q1 is the bottom for gross margins? And then Jensen, my question is for you. What is on your dashboard to give you the confidence that the strong demand can sustain into next year? And has DeepSeek and whatever innovations they came up with, has that changed that view in any way? Thank you.

Colette M. Kress -- Chief Financial Officer, Executive Vice President

Well, let me first take the first part of the question there regarding the gross margin. During our Blackwell ramp, our gross margins will be in the low 70s. At this point, we are focusing on expediting our manufacturing, expediting our manufacturing to make sure that we can provide to customers as soon as possible. Our Blackwell is fully round.

And once it does -- I'm sorry, once our Blackwell fully rounds, we can improve our cost and our gross margin. So, we expect to probably be in the mid-70s later this year. Walking through what you heard Jensen speak about the systems and their complexity, they are customizable in some cases. They've got multiple networking options.

They have liquid-cooled and water-cooled. So, we know there is an opportunity for us to improve these gross margins going forward. But right now, we are going to focus on getting the manufacturing complete and to our customers as soon as possible.

Jensen Huang -- President and Chief Executive Officer

We know several things, Vivek. We have a fairly good line of sight of the amount of capital investment that data centers are building out toward. We know that going forward, the vast majority of software is going to be based on machine learning. And so, accelerated computing and generative AI, reasoning AI are going to be the type of architecture you want in your data center.

We have, of course, forecasts and plans from our top partners. And we also know that there are many innovative, really exciting start-ups that are still coming online as new opportunities for developing the next breakthroughs in AI, whether it's agentic AIs, reasoning AI, or physical AIs. The number of start-ups are still quite vibrant and each one of them needs a fair amount of computing infrastructure. And so, I think the -- whether it's the near-term signals or the midterm signals, near-term signals, of course, are POs and forecasts and things like that.

Midterm signals would be the level of infrastructure and capex scale-out compared to previous years. And then the long-term signals has to do with the fact that we know fundamentally software has changed from hand-coding that runs on CPUs to machine learning and AI-based software that runs on GPUs and accelerated computing systems. And so, we have a fairly good sense that this is the future of software. And then maybe as you roll it out, another way to think about that is we've really only tapped consumer AI and search and some amount of consumer generative AI, advertising, recommenders, kind of the early days of software.

The next wave is coming, agentic AI for enterprise, physical AI for robotics, and sovereign AI as different regions build out their AI for their own ecosystems. And so, each one of these are barely off the ground, and we can see them. We can see them because, obviously, we're in the center of much of this development and we can see great activity happening in all these different places and these will happen. So, near term, midterm, long term.

Operator

Your next question comes from the line of Harlan Sur with JPMorgan. Please go ahead.

Harlan Sur -- Analyst

Yeah. Good afternoon. Thanks for taking my question. Your next-generation Blackwell Ultra is set to launch in the second half of this year, in line with the team's annual product cadence.

Jensen, can you help us understand the demand dynamics for Ultra given that you'll still be ramping the current generation Blackwell solutions? How do your customers and the supply chain also manage the simultaneous ramps of these two products? And is the team still on track to execute Blackwell Ultra in the second half of this year?

Jensen Huang -- President and Chief Executive Officer

Yes. Blackwell Ultra is second half. As you know, the first Blackwell was we had a hiccup that probably cost us a couple of months. We're fully recovered, of course.

The team did an amazing job recovering and all of our supply chain partners and just so many people helped us recover at the speed of light. And so, now we've successfully ramped production of Blackwell. But that doesn't stop the next train. The next train is on an annual rhythm and Blackwell Ultra with new networking, new memories, and of course, new processors, and all of that is coming online.

We have been working with all of our partners and customers, laying this out. They have all of the necessary information, and we'll work with everybody to do the proper transition. This time between Blackwell and Blackwell Ultra, the system architecture is exactly the same. It's

a lot harder going from Hopper to Blackwell because we went from an NVLink 8 system to an NVLink 72-based system.

So, the chassis, the architecture of the system, the hardware, the power delivery, all of that had to change. This was quite a challenging transition. But the next transition will slot right in Blackwell Ultra will slot right in. We've also already revealed and been working very closely with all of our partners on the click after that.

And the click after that is called Vera Rubin and all of our partners are getting up to speed on the transition of that and so preparing for that transition. And again, we're going to provide a big, huge step-up. And so, come to GTC, and I'll talk to you about Blackwell Ultra, Vera Rubin and then show you what we place after that. Really exciting new products to come to GTC piece.

Operator

Your next question comes from the line of Timothy Arcuri with UBS. Please go ahead.

Timothy Arcuri -- Analyst

Thanks a lot. Jensen, we heard a lot about custom ASICs. Can you kind of speak to the balance between customer ASIC and merchant GPU? We hear about some of these heterogeneous superclusters to use both GPU and ASIC. Is that something customers are planning on building? Or will these infrastructures remain fairly distinct? Thanks.

Jensen Huang -- President and Chief Executive Officer

Well, we built very different things than ASICs, in some ways, completely different in some areas we intercept. We're different in several ways. One, NVIDIA'S architecture is general whether you're -- you've optimized for unaggressive models or diffusion-based models or vision-based models or multimodal models, or text models. We're great in all of it.

We're great on all of it because our software stack is so -- our architecture is sensible. Our software stack ecosystem is so rich that we're the initial target of most exciting innovations and algorithms. And so, by definition, we're much, much more general than narrow. We're also really good from the end-to-end from data processing, the curation of the training data, to the training of the data, of course, to reinforcement learning used in post-training, all the way to inference with tough time scaling.

So, we're general, we're end-to-end, and we're everywhere. And because we're not in just one cloud, we're in every cloud, we could be on-prem. We could be in a robot. Our architecture is much more accessible and a great target initial target for anybody who's starting up a new company.

And so, we're everywhere. And the third thing I would say is that our performance in our rhythm is so incredibly fast. Remember that these data centers are always fixed in size. They're fixed in size or they're fixing power.

And if our performance per watt is anywhere from 2x to 4x to 8x, which is not unusual, it translates directly to revenues. And so, if you have a 100-megawatt data center, if the performance or the throughput in that 100-megawatt or the gigawatt data center is four times or eight times higher, your revenues for that gigawatt data center is eight times higher. And the reason that is so different than data centers of the past is because AI factories are directly monetizable through its tokens generated. And so, the token throughput of our architecture being so incredibly fast is just incredibly valuable to all of the companies that are building these things for revenue generation reasons and capturing the fast ROI.

And so, I think the third reason is performance. And then the last thing that I would say is

the software stack is incredibly hard. Building an ASIC is no different than what we do. We build a new architecture.

And the ecosystem that sits on top of our architecture is 10 times more complex today than it was two years ago. And that's fairly obvious because the amount of software that the world is building on top of architecture is growing exponentially and AI is advancing very quickly. So, bringing that whole ecosystem on top of multiple chips is hard. And so, I would say that -- those four reasons.

And then finally, I will say this, just because the chip is designed doesn't mean it gets deployed. And you've seen this over and over again. There are a lot of chips that get built, but when the time comes, a business decision has to be made, and that business decision is about deploying a new engine, a new processor into a limited AI factory in size, in power, and in fine. And our technology is not only more advanced, more performance, it has much, much better software capability and very importantly, our ability to deploy is lightning fast.

And so, these things are enough for the faint of heart, as everybody knows now. And so, there's a lot of different reasons why we do well, why we win.

Operator

Your next question comes from the line of Ben Reitzes with Melius Research. Please go ahead.

Ben Reitzes -- Analyst

Yeah. Hi. Ben Reitzes here. Hey, thanks a lot for the question.

Jensen, it's a geography-related question. You did a great job explaining some of the demand underlying factors here on the strength. But U.S. was up about \$5 billion or so sequentially.

And I think there is a concern about whether U.S. can pick up the slack if there's regulations toward other geographies. And I was just wondering, as we go throughout the year, if this kind of surge in the U.S. continues and it's going to be -- whether that's OK.

And if that underlies your growth rate, how can you keep growing so fast with this mix shift toward the U.S.? Your guidance looks like China is probably up sequentially. So, just wondering if you could go through that dynamic and maybe collect can weigh in. Thanks a lot.

Jensen Huang -- President and Chief Executive Officer

China is approximately the same percentage as Q4 and as previous quarters. It's about half of what it was before the export control. But it's approximately the same in percentage. With respect to geographies, the takeaway is that AI is software.

It's modern software. It's incredible modern software, but it's modern software and AI has gone mainstream. AI is used in delivery services everywhere, shopping services everywhere. If you were to buy a quart of milk, it's delivered to you.

AI was involved. And so almost everything that a consumer service provides AIs at the core of it. Every student will use AI as a tutor, healthcare services use AI, financial services use AI. No fintech company will not use AI.

Every Fintech company will. Climate tech companies use AI. Mineral discovery now uses AI. The number of -- every higher education, every university uses AI and so I think it is fairly safe to say that AI has gone mainstream and that it's being integrated into every application.

And our hope is that, of course, the technology continues to advance safely and advance in a helpful way to society. And with that, we're -- I do believe that we're at the beginning of

this new transition. And what I mean by that in the beginning is, remember, behind us has been decades of data centers and decades of computers that have been built. And they've been built for a world of hand coding and general-purpose computing and CPUs and so on and so forth.

And going forward, I think it's fairly safe to say that world is going to be almost all software to be infused with AI. All software and all services will be based on -- ultimately, based on machine learning, the data flywheel is going to be part of improving software and services and that the future computers will be accelerated, the future computers will be based on AI. And we're really two years into that journey. And in modernizing computers that have taken decades to build out.

And so, I'm fairly sure that we're in the beginning of this new era. And then lastly, no technology has ever had the opportunity to address a larger part of the world's GDP than AI. No software tool ever has. And so, this is now a software tool that can address a much larger part of the world's GDP more than any time in history.

And so, the way we think about growth, and the way we think about whether something is big or small has to be in the context of that. And when you take a step back and look at it from that perspective, we're really just in the beginning.

Operator

Your next question comes from the line of Aaron Rakers with Wells Fargo. Please go ahead. Aaron, your line is open. Your next question comes from Mark Lipacis with Evercore ISI.

Please go ahead.

Mark Lipacis -- Analyst

Hi. It's Mark. Thanks for taking the question. I had a clarification and a question.

Colette up for the clarification. Did you say that enterprise within the data center grew 2x year on year for the January quarter? And if so, does that -- would that make it the fast faster growing than the hyperscalers? And then, Jensen, for you, the question, hyperscalers are the biggest purchasers of your solutions, but they buy equipment for both internal and external workloads, external workflows being cloud services that enterprise is used. So, the question is, can you give us a sense of how that hyperscaler spend splits between that external workload and internal? And as these new AI workflows and applications come up, would you expect enterprises to become a larger part of that consumption mix? And does that impact how you develop your service, your ecosystem? Thank you.

Colette M. Kress -- Chief Financial Officer, Executive Vice President

Sure. Thanks for the question regarding our Enterprise business. Yes, it grew 2x and very similar to what we were seeing with our large CSPs. Keep in mind, these are both important areas to understand.

Working with the CSPs can be working on large language models, can be working on inference in their own work. But keep in mind, that is also where the enterprises are servicing. Your enterprises are both with your CSPs as well as in terms of building on their own. They're both growing quite well.

Jensen Huang -- President and Chief Executive Officer

The CSPs are about half of our business. And the CSPs have internal consumption and external consumption, as you say. And we're using -- of course, used for internal consumption. We work very closely with all of them to optimize workloads that are internal to them because they have a large infrastructure of NVIDIA gear that they could take advantage of.

And the fact that we could be used for AI on the one hand, video processing on the other hand, data processing like Spark, we're fungible. And so, the useful life of our infrastructure is much better. If the useful life is much longer, then the TCO is also lower. And so, the second part is how do we see the growth of enterprise or not CSPs, if you will, going forward? And the answer is, I believe, long term, it is by far larger and the reason for that is because if you look at the computer industry today and what is not served by the computer industry is largely industrial.

So, let me give you an example. When we say enterprise, and let's use the car company as an example because they make both soft things and hard things. And so, in the case of a car company, the employees will be what we call enterprise and agentic AI and software planning systems and tools, and we have some really exciting things to share with you guys at GTC, build agentic systems are for employees to make employees more productive to design to market plan to operate their company. That's agentic AI.

On the other hand, the cars that they manufacture also need AI. They need an AI system that trains the cars, treats this entire giant fleet of cars. And today, there's 1 billion cars on the road. Someday, there will be 1 billion cars on the road, and every single one of those cars will be robotic cars, and they'll all be collecting data, and we'll be improving them using an AI factory.

Whereas they have a car factory today, in the future, they'll have a car factory and an AI factory. And then inside the car itself is a robotic system. And so, as you can see, there are three computers involved and there's the computer that helps the people. There's the computer that build the AI for the machineries that could be, of course, could be a tractor, it could be a lawn mower.

It could be a human or robot that's being developed today. It could be a building. It could be a warehouse. These physical systems require new type of AI we call physical AI.

They can't just understand the meaning of words and languages, but they have to understand the meaning of the world, friction and inertia, object permanence, and cause and effect. And all of those type of things that are common sense to you and I, but AIs have to go learn those physical effects. So, we call that physical AI. That whole part of using agentic AI to revolutionize the way we work inside companies, that's just starting.

This is now the beginning of the agentic AI era, and you hear a lot of people talking about it, and we got some really great things going on. And then there's the physical AI after that, and then there are robotic systems after that. And so, these three computers are all brand new. And my sense is that long term, this will be by far the larger of a mall, which kind of makes sense.

The world's GDP is representing -- represented by either heavy industries or industrials and companies that are providing for those.

Operator

Your next question comes from the line of Aaron Rakers with Wells Fargo. Please go ahead.

Aaron Rakers -- Analyst

Yeah. Thanks for letting me back in. Jensen, I'm curious as we now approach the two-year anniversary of really the Hopper inflection that you saw in 2023 in GenAI in general. And when we think about the road map you have in front of us, how do you think about the infrastructure that's been deployed from a replacement cycle perspective and whether if it's GB300 or if it's the Rubin cycle where we start to see maybe some refresh opportunity? I'm just curious how you look at that.

Jensen Huang -- President and Chief Executive Officer

Yeah. I appreciate it. First of all, people are still using Voltas and Pascals, and Amperes. And the reason for that is because there are always things that because CUDA is so programmable you could use it Blackwell, one of the major use cases right now is data processing and data curation.

You find a circumstance that an AI model is not very good at. You present that circumstance to a vision language model, let's say, it's a car. You present that circumstance to a vision language model. The vision language model actually looks in the circumstances, said, this is what happened and I was very good at it.

You then take that response to the prompt, and you go and prompt an AI model to go find in your whole lake of data other circumstances like that, whatever that circumstance was. And then you use an AI to do domain randomization and generate a whole bunch of other examples. And then from that, you can go train the bottle. And so, you could use an peers to go and do data processing and data curation and machine learning-based search.

And then you create the training data set, which you then present to your Hopper systems for training. And so, each one of these architectures are completely -- they're all CUDA-compatible and so everything wants on everything. But if you have infrastructure in place, then you can put the less intensive workloads onto the installed base of the past. All of us are very well employed.

Operator

We have time for one more question, and that question comes from Atif Malik with Citi. Please go ahead.

Atif Malik -- Analyst

Hi. Thank you for taking my question. I have a follow-up question on gross margins for Colette. So, I understand there are many moving parts the Blackwell yields, NVLink 72, and Ethernet mix.

And you kind of tipped to the earlier question the April quarter is the bottom, but second half would have to ramp like 200 basis points per quarter to get to the mid-70s range that you're giving for the end of the fiscal year. And we still don't know much about tariff impact to broader semiconductor, so what kind of gives you the confidence in that trajectory in the back half of this year?

Colette M. Kress -- Chief Financial Officer, Executive Vice President

Yeah. Thanks for the question. Our gross margins, they're quite complex in terms of the material and everything that we put together in a Blackwell system, a tremendous amount of opportunity to look at a lot of different pieces of that on how we can better improve our gross margins over time. Remember, we have many different configurations as well on Blackwell that will be able to help us do that.

So, together, working after we get some of these really strong ramping completed for our customers, we can begin a lot of that work. If not, we're going to probably start as soon as possible if we can. If we can improve it in the short term, we will also do that. Tariff at this point, it's a little bit of an unknown it's an unknown until we understand further what the U.S.

government's plan is, both its timing, it's where, and how much. So, at this time, we are awaiting, but again, we would, of course, always follow export controls and/or tariffs in that

manner.

Operator

Ladies and gentlemen, that does conclude our question-and-answer session. I'm sorry.

Jensen Huang -- President and Chief Executive Officer

Thank you.

Colette M. Kress -- Chief Financial Officer, Executive Vice President

We are going to open up to Jensen. A couple of things.

Jensen Huang -- President and Chief Executive Officer

I just wanted to thank you. Thank you, Colette. The demand for Blackwell is extraordinary. AI is evolving beyond perception and generative AI into reasoning.

With resending AI, we're observing another scaling law, inference time or test time scaling, more computation. The more the model thinks the smarter the answer. Models like OpenAI, Grok 3, DeepSeek-R1 are reasoning models that apply inference time scaling. Reasoning models can consume 100x more compute.

Future reasoning models can consume much more compute. DeepSeek-R1 has ignited global enthusiasm. It's an excellent innovation. But even more importantly, it has open-sourced a world-class reasoning AI model.

Nearly every AI developer is applying R1 or chain of thought and reinforcement learning techniques like R1 to scale their model's performance. We now have three scaling laws, as I mentioned earlier, driving the demand for AI computing. The traditional scaling loss of AI remains intact. Foundation models are being enhanced with multimodality, and pretraining is still growing.

But it's no longer enough. We have two additional scaling dimensions: post-training scaling, where reinforcement learning, fine-tuning, model distillation require orders of magnitude more compute than pretraining alone; inference time scaling and reasoning where a single query and demand 100x more compute. We defined Blackwell for this moment, a single platform that can easily transition from pre-training, post-training, and test time scaling. Blackwell's FP4 transformer engine and NVLink 72 scale-up fabric and new software technologies led Blackwell process reasoning AI models 25 times faster than Hopper.

Blackwell in all of this configuration is in full production. Each Grace Blackwell NVLink 72 rack is an engineering marvel. 1.5 million components produced across 350 manufacturing sites by nearly 100,000 factory operators. AI is advancing at light speed.

We're at the beginning of reasoning AI and inference time scaling. But we're just at the start of the age of AI, multimodal AIs, enterprise AI sovereign AI and physical AI are right around the corner. We will grow strongly in 2025. Going forward, data centers will dedicate most of capex to accelerated computing and AI.

Data centers will increasingly become AI factories, and every company will have either rented or self-operated. I want to thank all of you for joining us today. Come join us at GTC in a couple of weeks. We're going to be talking about Blackwell Ultra, Rubin, and other new computing, networking, reasoning AI, physical AI products, and a whole bunch more.

Thank you.

Operator

[Operator signoff]

Duration: 0 minutes

Call participants:

Stewart Stecker -- Senior Director, Investor Relations

Colette M. Kress -- Chief Financial Officer, Executive Vice President

C.J. Muse -- Analyst

Jensen Huang -- President and Chief Executive Officer

Joe Moore -- JPMorgan Chase and Company -- Analyst

Vivek Arya -- Analyst

Colette Kress -- Chief Financial Officer, Executive Vice President

Harlan Sur -- Analyst

Timothy Arcuri -- Analyst

Ben Reitzes -- Analyst

Mark Lipacis -- Analyst

Aaron Rakers -- Analyst

Atif Malik -- Analyst

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